Asiri Sandakelum

Final Year Mechanical Engineering Undergraduate, University of Moratuwa

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Education

University of Moratuwa Moratuwa, Sri Lanka

B.Sc. Engineering (Hons.) in Mechanical Engineering; SGPA: 3.3/4.0 *2022 - Present*

- *Manufacturing Processes, PLC, Embedded Systems, Machine Design, 3D Modeling*

HK/Denike National School Rikillagaskada, Sri Lanka

GCE Advanced Level 2020 - Physical Science Stream; Z-Score 2.4781, Island Rank 175th *2018 - 2021*

- *Combined Mathematics, Physics, Chemistry*

Experience

Future Fibres Lanka (Pvt) Ltd Biyagama EPZ, Sri Lanka

Mechanical Engineering Intern *Dec 2024 – May 2025*

- Gained hands-on exposure to advanced composite manufacturing processes, including pultrusion, winding, braiding, and PU molding for carbon fiber rigging systems exported globally.
- Redesigned a **Bobbin Winding Tensioner** using SolidWorks, resolving roller misalignment; improved yarn tension consistency and reduced breakage after fabrication and testing.
- Conducted a carbon dust filtration case study for the AeroSIX plate cutting area; designed a sealed HEPA filtration room, significantly improving air quality and worker safety.
- Developed an **IoT-based Temperature & Humidity Monitoring System** using ESP32, DHT22, and DS18B20 sensors with real-time cloud logging via ThingSpeak.
- Supported preventive maintenance scheduling, CMMS implementation, AutoCAD layout drafting (floor plans, piping, drainage), and 5S standardization activities.

Projects

Active Knee Exoskeleton for Osteoarthritis Treatment *Ongoing*

- Designed a wearable active exoskeleton to reduce Knee Adduction Moment (KAM) for mild-to-moderate knee osteoarthritis patients. Optimized a polycentric 5-Bar Linkage via Genetic Algorithm (GA) to mimic the knee's Instantaneous Center of Rotation (ICR), achieving a tracking error of 0.47 mm.
- Features remote cable-driven actuation, FSM-based gait detection, and impedance control. Target assistive torque: 25 Nm.
- **Tools:** MATLAB, Genetic Algorithm, SolidWorks

Enhancing Operational Throughput – Case Study at Abans Auto (Pvt) Ltd *Ongoing*

- Analyzed Hero Motorcycle Assembly Plant operations; identified bottlenecks in PDI/QC (75 min/unit) and material handling waste (90 min/batch). Proposed facility layout optimization, drive-through workstation configuration, dedicated functional test track, and strategic safety stock management.

Below-Knee Prosthetic Leg Fatigue Load Testing Mechanism *Dec 2025*

- Designed a fully mechanical walking mechanism simulating the human gait cycle (heel-strike, stance, toe-off, swing) for fatigue load testing of prosthetic legs, withstanding vertical loads up to 100 kg.
- Applied weighted decision matrices to select the optimal guided frame crank-link mechanism.
- **Tools:** SolidWorks, DFMA Principles, 3D Printing (PLA)

STEM Development & Experiment Kit – Feasibility Study *Nov 2025*

- Comprehensive feasibility study (market, technical, financial) for an eco-friendly STEM kit for children.
- **Tools:** MS Excel, Power Point, Gantt Charts, Risk Matrices

Autonomous Robot – Path Following & Obstacle Avoidance *Ongoing*

- Designed an autonomous robot for line following, wall following, and obstacle avoidance at 4-way junctions with mobile app control and dead-zone recovery.
- **Tools:** Arduino, Mobile App, Sensors

Gearbox Design – Combat Robot Weapons *Nov 2024*

- Designed a high-torque, multi-speed gearbox for vertical and horizontal spinner combat robots; performed gear selection, shaft design, and power transmission calculations.
- **Tools:** Solid Edge, Machine Design Principles, CAD

Optimization of Production Line – Aruna Tea Factory *Nov 2024*

- Identified bottlenecks in withering and packing sections; proposed scheduling optimization and equipment modifications through simulation-based analysis.

Reverse Engineering – Bicycle Disc Brake System *May 2024*

- Performed material identification (burn tests, density tests, Rockwell hardness) and developed optimized 3D models and simulations for improved brake performance.
- **Tools:** Fusion 360, FEA, Generative Design

Smart Medibox – IoT Embedded Project *Aug 2024*

- Designed an IoT system to monitor and control environmental conditions with real-time dashboard visualisation.
- **Tools:** C++, Node-RED, MQTT, Wokwi

Autonomous Car Wash Robot (Mechatronics Group Project) *Jul 2023*

- Designed mechanical system and integrated Arduino Mega, ESP module, and mobile app control.
- **Tools:** SolidWorks, Arduino, ESP Module, MIT App Inventor

RC Combat Robot *Jan 2024*

- Designed and manufactured an RC combat robot for the UWU Death Racing Competition.
- **Tools:** Arduino, FlySky Transmitter

Leadership & Volunteering

Department of Mechanical Engineering, University of Moratuwa
Student Representative *Final Academic Year*

- Acted as the primary liaison between Department administration and the student body; negotiated academic schedules and resolved administrative grievances to ensure student welfare.
- Spearheaded coordination of departmental career fairs and industry networking sessions, connecting students with leading engineering firms.
- Led planning and execution of extracurricular events and professional development workshops, managing cross-functional student committees and budget allocations.
- Developed a centralized communication framework for department-wide updates, achieving 100% information reach and increasing student engagement.

IEEE Robotics and Automation Society (RAS), University of Moratuwa
Vice Chairman *Nov 2024 – Oct 2025*

- Spearheaded executive decision-making and strategic planning for one of the university's most active technical chapters.
- Mentored junior committees in event execution and technical project management.

Co-Chair, ***BatTalk 2.0*** *Mar 2024 – Oct 2024*

- Directed planning and execution of a flagship technical talk series on AI/ML job dynamics, coordinating with industry experts and university faculty.

Programme & Logistics Committee Lead, ***MORA NexGen 1.0*** *Jan 2024 – Mar 2024*

- Led a team to manage end-to-end event logistics for a high-profile robotics competition targeting school students.

Sasnaka Sansada Foundation
District Coordinator*, Nuwara Eliya* *Mar 2024 – Dec 2024*

- Oversaw district-level operations and volunteer recruitment for community outreach programs including provincial events, meetups, and seminar series.

Ganitha Sawiya Coordinator *May 2022 – May 2023*

- Organized over 50 mathematics seminars and educational workshops for school students in the Nuwara Eliya District to improve regional pass rates.

Volunteer *Jun 2021 – Apr 2022*

EXMO 2023, University of Moratuwa
Exhibit Coordinator *Apr 2023 – Jul 2023*

- Curated and managed mechanical engineering exhibits for the university's premier national engineering exhibition, handling space allocation and technical safety for thousands of visitors.

Skills Summary

CAD : SolidWorks, Fusion 360, Solid Edge, AutoCAD

Simulation & Analysis: ANSYS, MATLAB , FEA, Genetic Algorithm

Programming: Python , C++ , Arduino IDE

IoT & Embedded: ESP32, DHT22, DS18B20, ThingSpeak, Node-RED, MQTT, Teensy 4.1

Manufacturing: Composite manufacturing (pultrusion, braiding, winding, molding), CNC machining, sheet metal fabrication, welding

Software: MS Office, Photoshop, DaVinci Resolve, After Effects

Languages: English (professional proficiency), Sinhala (native proficiency)

Soft Skills: Leadership, Teamwork, Public Speaking, Technical Documentation

Sports: Cricket, Volleyball

References

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